PRACTISE QUESTIONS

(Text Book : Dietel & Dietel)

Chapter 3: Introduction to Classes, Objects and Strings

Exercise Problems:

3.11 (Modifying Class GradeBook) Modify class GradeBook (Figs. 3.11–3.12) as follows:
a) Include a second string data member that represents the course instructor's name.
b) Provide a set function to change the instructor's name and a get function to retrieve it.
c) Modify the constructor to specify course name and instructor name parameters.
d) Modify function **displayMessage** to output the welcome message and course name, then the string "This course is presented by: " followed by the instructor's name.

Use your modified class in a test program that demonstrates the class's new capabilities.

3.12 (Account Class) Create an Account class that a bank might use to represent customers' bank accounts. Include a data member of type int to represent the account balance.

Provide a constructor that receives an initial balance and uses it to initialize
the data member. The constructor should validate the initial balance to ensure that it's greater than or equal to 0. If not, set the balance to 0 and display an error message indicating that the initial balance was invalid.

Provide three member functions.

- Member function **credit** should add an amount to the current balance.
- Member function **debit** should withdraw money from the Account and ensure that the debit amount does not exceed the Account's balance. If it does, the balance should be left unchanged and the function should print a message indicating "Debit amount exceeded account balance."
- Member function **getBalance** should return the current balance.

Create a program that creates two Account objects and tests the member functions of class Account.

3.13 (Invoice Class) Create a class called Invoice that a hardware store might use to represent an invoice for an item sold at the store. An Invoice should include four data members—a part number (type string), a part description (type string), a quantity of the item being purchased (type int) and a price per item (type int).

Your class should have a constructor that initializes the four data members. Provide a set and a get function for each data member. In addition, provide a member function named **getInvoiceAmount** that calculates the invoice amount (i.e., multiplies the quantity by the price per item), then returns the amount as an int value. If the quantity is not positive, it should be set to 0. If the price per item is not positive, it should be set to 0. Write a test program that demonstrates class Invoice's capabilities.

3.14 (Employee Class) Create a class called Employee that includes three pieces of information as data members—a first name (type string), a last name (type string) and a monthly salary (type int). Your class should have a constructor that initializes the three data members. Provide a set and a get function for each data member. If the monthly salary is not positive, set it to 0. Write a test program that demonstrates class Employee's capabilities. Create two Employee objects and display each object's yearly salary. Then give each Employee a 10 percent raise and display each Employee's yearly salary again.

3.15 (Date Class) Create a class called Date that includes three pieces of information as data members—a month (type int), a day (type int) and a year (type int). Your class should have a constructor with three parameters that uses the parameters to initialize the three data members. For the purpose of this exercise, assume that the values provided for the year and day are correct, but ensure that the month value is in the range 1–12; if it isn't, set the month to 1. Provide a set and a get function for each data member. Provide a member function displayDate that displays the month, day and year separated by forward slashes (/). Write a test program that demonstrates class Date's capabilities.

Chapter 9: Classes (A deeper Look) Part 1

Book Examples to Cover: Example no 9.1 - Time class defination (Page 381), Example no 9.2 - Time class member-function definitions (Page 383), Example no 9.3 - (Page 386), Example no 9.4 - (Page 389), Example no 9.5 - (Page 391), Example no 9.6 - (Page 391- 392), Example no 9.7 - (Page 392-393), Example no 9.8 - (Page 394), Example no 9.9 - (Page 395), Example no 9.11 - (Page 400), Example no 9.12 - (Page 400-401), Example no 9.14 - (Page 403), Example no 9.15 - (Page 404), Example no 9.16 - (Page 405), Example no 9.17 - (Page 406), Example no 9.18 - (Page 406), Example no 9.19 - (Page 407).

Exercise Problems for Practise:
Page no 411 - Question no 9.5, 9.6, 9.7, 9.8, 9.9, 9.11, 9.14.